

Quesito A (Solo Esame completo)

Assegnato il manipolatore in figura, con lunghezze dei bracci $a_1 = a_2 = 1$, $a_3 = 0.5$, e posto $a_0 = 1$:

1. Si fissino le terne solidali con ciascun braccio seguendo la convenzione di Denavit-Hartenberg, giustificando adeguatamente la scelta;
2. Si calcoli la funzione cinematica diretta;
3. Si calcoli lo Jacobiano geometrico;
4. Si calcolino posizione e orientamento delle terne solidali con ciascun braccio, rispetto alla terna base, quando il vettore delle variabili di giunto assume il valore $q = \left[\frac{\pi}{2} \quad 0.5 \quad 0.2\right]^T$.
5. Si individui il vettore delle variabili di giunto corrispondente alla posizione dell'origine della terna dell'organo terminale data da $[-1, 1, 0.5]$. Si se determini il punto $[1, -1.5, 0.6]$ appartenga o meno allo spazio di lavoro del manipolatore.
6. Con riferimento alla configurazione del punto 4., si determini la velocità di ciascuna terna rispetto al riferimento fisso assumendo una velocità dei giunti data da $\dot{q} = [0 \quad 1 \quad 1]^T$. Si commenti il risultato.

Quesito B (Esame completo ed Esonero)

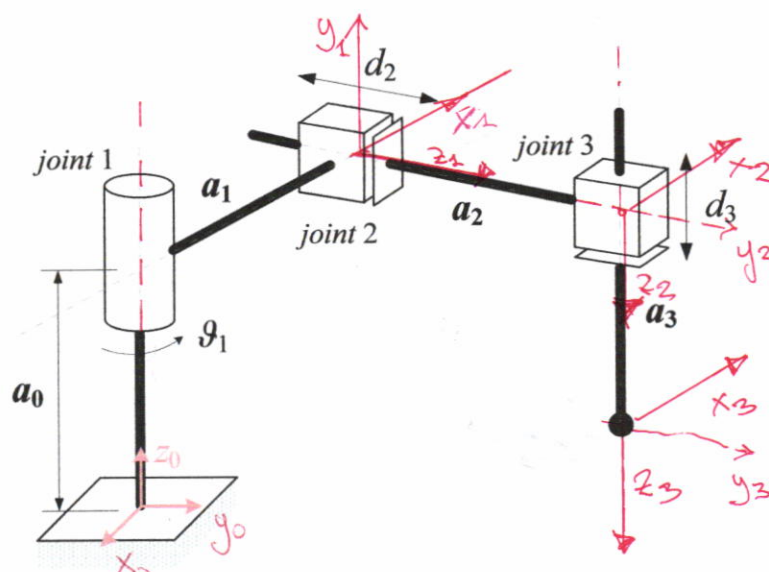
Si definisca la traiettoria per la variabile di un giunto (si caratterizzi matematicamente la legge oraria in termini di posizione $q(t)$, velocità $\dot{q}(t)$, accelerazione $\ddot{q}(t)$) adoperando tratti parabolici e lineari con passaggio in prossimità dei punti di via $q = 0, 2\pi, \pi/2$ negli istanti $t = 0, 2, 3$, con durate dei tratti parabolici $\Delta t' = 0.2, 0.2, 0.4$.

Quesito C (Esame completo ed Esonero)

Adottando un linguaggio sintetico ed appropriato, si discuta del controllo indipendente ai giunti, con retroazione di posizione e controllore PI. In particolare, si illustri:

- In cosa consiste e quali sono i principi alla base del controllo indipendente ai giunti, e sotto quali ipotesi sia possibile utilizzare uno schema di controllo indipendente ai giunti;
- Lo schema di controllo in feedback per il singolo giunto con controllore di posizione;
- Un metodo di taratura dei parametri del regolatore che consenta di ottenere una dinamica del secondo ordine sottosmorzata in anello chiuso;
- Uno schema di compensazione in avanti decentralizzata.

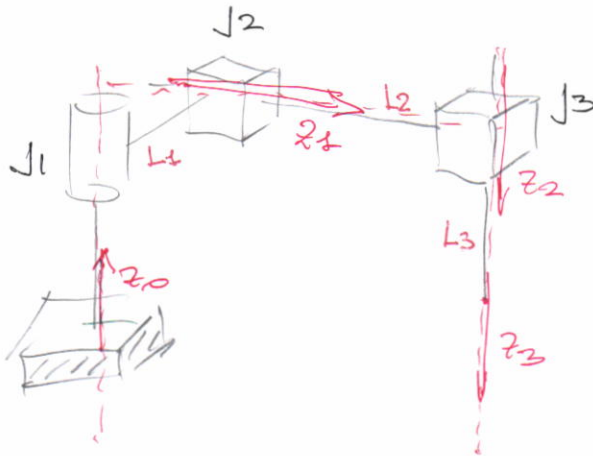
Infine, si discuta della compensazione dei termini di accoppiamento mediante uno schema di controllo a coppia precalcolata.



A.1

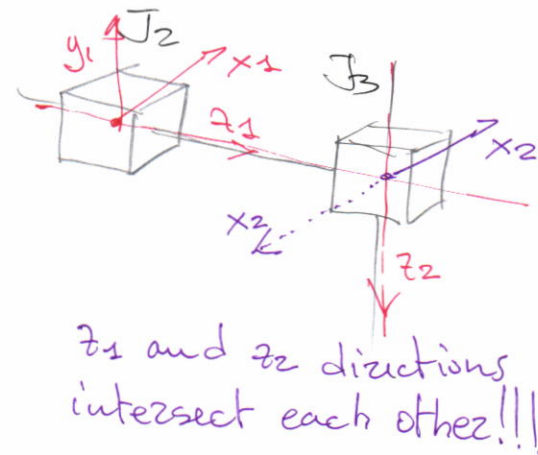
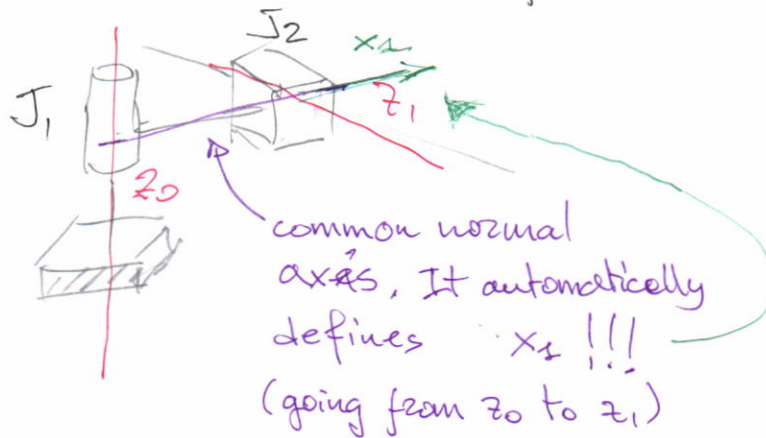
SETTING THE LINK FRAMES ACCORDING TO D-H RULES

- Identify joints axes and set z_i axes accordingly

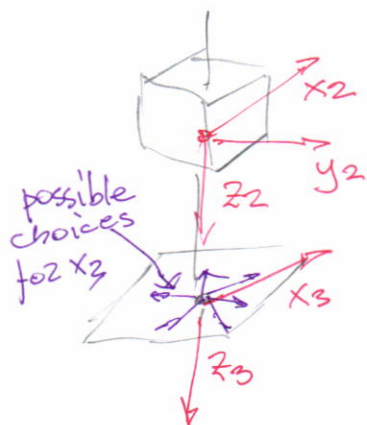


!! z_3 direction is not uniquely defined, because a joint #4 does not exist! It is convenient to set the same direction as z_2 , so that $a_3=0$ and $\alpha_3=0$

- Identify the common normal to axes z_{i-1} and z_i . It determines the direction of x_i



Since the common normal degenerates in a point, x_2 is only constrained to be on the normal axis to plane defined by z_1 and z_2 directions



Since z_2 and z_3 are parallel, we can identify infinite common normal directions, lying on a plane. It is convenient to choose x_3 parallel to x_2 , resulting in $\theta_3=0$

A.1 & A.2

$$a_1 = a_2 = 1 ; a_3 = 0.5 ; a_0 = 1$$

$$A_{i-1}^i = \begin{bmatrix} c_{\alpha_i} & -s_{\alpha_i} c_{\alpha_i} & s_{\alpha_i} s_{\alpha_i} & a_i c_{\alpha_i} \\ s_{\alpha_i} & c_{\alpha_i} c_{\alpha_i} & -c_{\alpha_i} s_{\alpha_i} & a_i s_{\alpha_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

LINK	a_i	α_i	d_i	θ_i
#1	a_1	$\pi/2$	a_0	θ_1
#2	$\emptyset$	$\pi/2$	d_2	$\emptyset$
#3	$\emptyset$	$\emptyset$	d_3	$\emptyset$

$$A_1^0 = \begin{bmatrix} c_{\alpha_1} & 0 & \begin{bmatrix} z_1 \\ s_{\alpha_1} \end{bmatrix} & \begin{bmatrix} p_1 \\ a_1 c_{\alpha_1} \end{bmatrix} \\ s_{\alpha_1} & 0 & \begin{bmatrix} -c_{\alpha_1} \end{bmatrix} & \begin{bmatrix} a_1 s_{\alpha_1} \end{bmatrix} \\ 0 & 1 & \begin{bmatrix} 0 \end{bmatrix} & \begin{bmatrix} d_0 \end{bmatrix} \\ 0 & 0 & \begin{bmatrix} 0 \end{bmatrix} & \begin{bmatrix} 1 \end{bmatrix} \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$a_0 = a_1 = 1$$

$$A_2^0 = \begin{bmatrix} c_{\alpha_1} & 0 & s_{\alpha_1} & c_{\alpha_1} \\ s_{\alpha_1} & 0 & -c_{\alpha_1} & s_{\alpha_1} \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T(q) = A_1^0 A_2^1 A_3^2 = \begin{bmatrix} c_{\alpha_1} & 0 & s_{\alpha_1} & a_1 c_{\alpha_1} \\ s_{\alpha_1} & 0 & -c_{\alpha_1} & a_1 s_{\alpha_1} \\ 0 & 1 & 0 & d_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\equiv A_3^0$$

$$T(q) = \begin{bmatrix} c_{\alpha_1} & s_{\alpha_1} & \begin{bmatrix} z_2 \\ 0 \end{bmatrix} & \begin{bmatrix} p_2 \\ d_2 s_{\alpha_1} + a_1 c_{\alpha_1} \end{bmatrix} \\ s_{\alpha_1} & -c_{\alpha_1} & \begin{bmatrix} 0 \end{bmatrix} & \begin{bmatrix} -d_2 c_{\alpha_1} + a_1 s_{\alpha_1} \end{bmatrix} \\ 0 & 0 & \begin{bmatrix} -1 \end{bmatrix} & \begin{bmatrix} a_0 \end{bmatrix} \\ 0 & 0 & \begin{bmatrix} 0 \end{bmatrix} & \begin{bmatrix} 1 \end{bmatrix} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$\Rightarrow$

A_2^0

$$T(q) = \begin{bmatrix} \cos q_1 & \sin q_1 & \begin{bmatrix} z_3 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} & \begin{bmatrix} d_2 \sin q_1 + a_1 \cos q_1 \\ -d_2 \cos q_1 + a_1 \sin q_1 \\ -d_3 + a_0 \\ 1 \end{bmatrix} \end{bmatrix}$$

2

A.3

$$J = \begin{bmatrix} z_0 \wedge (p - p_0) & z_1 & z_2 \\ z_0 & 0 & 0 \end{bmatrix}$$

$$p = \begin{bmatrix} d_2 \sin q_1 + a_1 \cos q_1 \\ a_1 \sin q_1 - d_2 \cos q_1 \\ a_0 - d_3 \end{bmatrix}; \quad p_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \quad z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad z_1 = \begin{bmatrix} \sin q_1 \\ -\cos q_1 \\ 0 \end{bmatrix}; \quad z_2 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$z_0 \wedge (p - p_0) = \begin{bmatrix} i & j & k \\ 0 & 0 & 1 \\ a_1 \cos q_1 + d_2 \sin q_1 & a_1 \sin q_1 - d_2 \cos q_1 & a_0 - d_3 \end{bmatrix} =$$

$$= [d_2 \cos q_1 - a_1 \sin q_1] \underline{i} + [a_1 \cos q_1 + d_2 \sin q_1] \underline{j} + 0 \underline{k}$$

$$J = \begin{bmatrix} d_2 \cos q_1 - a_1 \sin q_1 & \sin q_1 & 0 \\ a_1 \cos q_1 + d_2 \sin q_1 & -\cos q_1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

A.4

To be determined: R_1^0, R_2^0, R_3^0 and $p_1, p_2, p_3 = p$ for:

$$q = \begin{bmatrix} \pi/2 & 0.5 & 0.2 \end{bmatrix} \text{ and } a_0 = a_1 = 1$$

θ_1 d_2 d_3

First link frame position and orientation, given q :

$$A_1^0 = \begin{bmatrix} \cos \theta_1 & 0 & \sin \theta_1 & d_1 \cos \theta_1 \\ \sin \theta_1 & 0 & -\cos \theta_1 & a_1 \sin \theta_1 \\ 0 & 1 & 0 & a_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

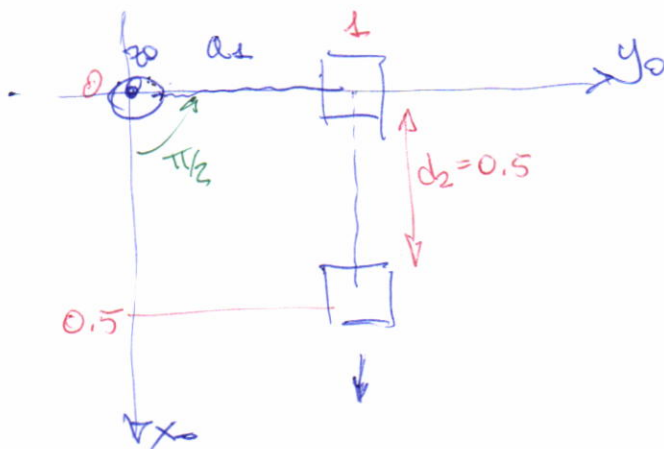
R_1^0 a_1 p_1

$$A_2^0 = \begin{bmatrix} \cos \theta_1 & \sin \theta_1 & 0 & d_2 \cos \theta_1 + a_1 \cos \theta_1 \\ \sin \theta_1 & -\cos \theta_1 & 0 & -d_2 \sin \theta_1 + a_1 \sin \theta_1 \\ 0 & 0 & -1 & a_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R_2^0 p_2

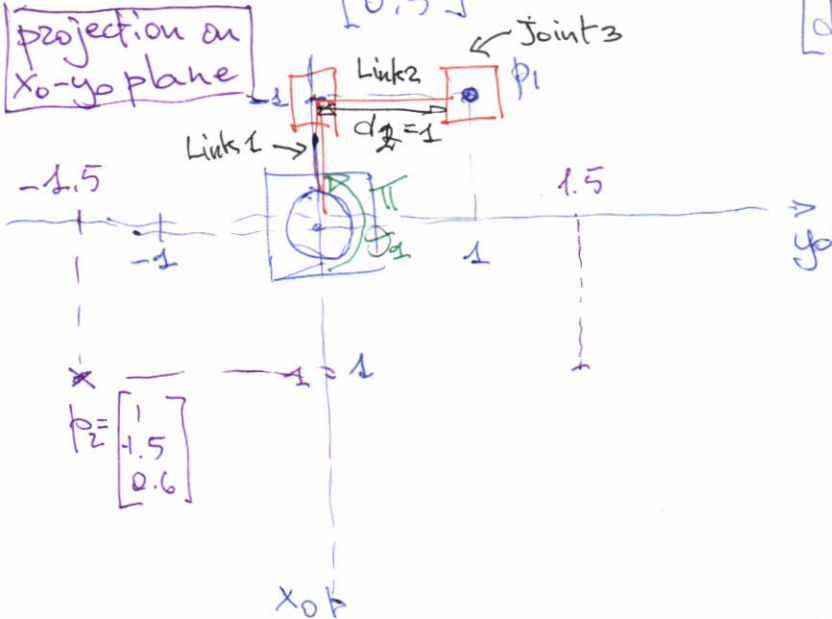
$$A_3^0 = \begin{bmatrix} \cos \theta_1 & \sin \theta_1 & 0 & d_2 \cos \theta_1 + a_1 \cos \theta_1 \\ \sin \theta_1 & -\cos \theta_1 & 0 & -d_2 \sin \theta_1 + a_1 \sin \theta_1 \\ 0 & 0 & -1 & -d_3 + a_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0.8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R_3^0 p_3



Given $p_1 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \end{bmatrix}$, find $q = \begin{bmatrix} \theta_1 \\ d_2 \\ d_3 \end{bmatrix}$ to obtain it

projection on x_0-y_0 plane



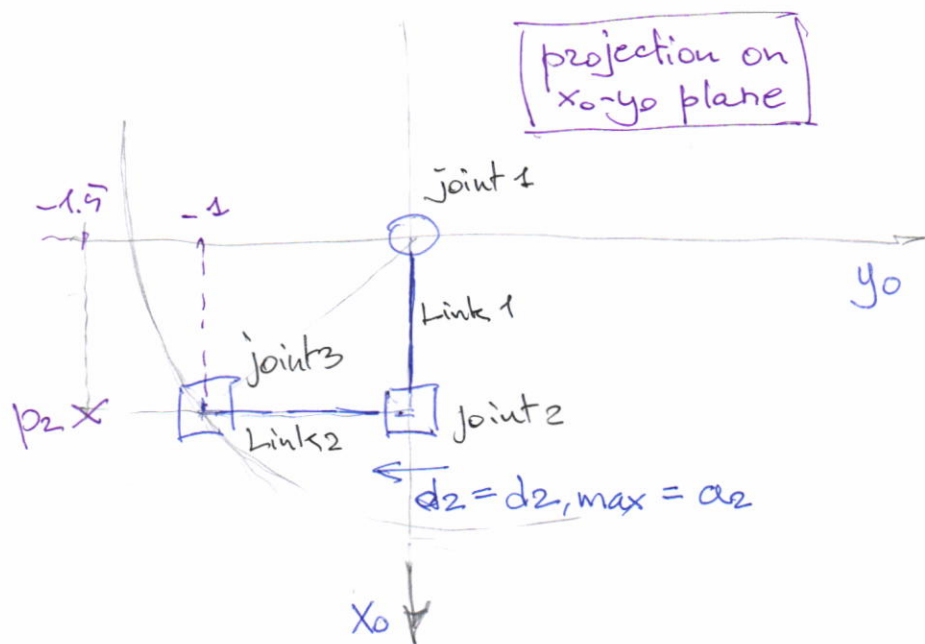
$$\theta_1 = \pi$$

$$d_2 = 1$$

$$d_3 = 0.5$$

$$q = \begin{bmatrix} \pi \\ 1 \\ 0.5 \end{bmatrix}$$

According to the choice made for reference frames, p_2 does not belong to the working space of the manipulator, because p_2 coordinate overcomes the length of link 2



A.6

Given $\dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{d}_2 \\ \dot{d}_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

and $q = [\pi/2 \ 0.5 \ 0.2]^T$
find $\dot{p}_1, \dot{p}_2, \dot{p}_3, \omega_1, \omega_2, \omega_3$

$$J_{L1} = \begin{bmatrix} z_0 \wedge (p_1 - p_0) & 0 & 0 \\ z_0 & 0 & 0 \end{bmatrix} \quad p_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$z_0 \wedge (p_1 - p_0) = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = -\hat{i} + 0\hat{j} + 0\hat{k}$$

$$J_{L1}(q) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad v_1 = J_{L1} \cdot \dot{q} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{matrix} \dot{p}_1 \\ \omega_1 \end{matrix}$$

Linear velocity of p_1 is null because joint #1 is not moving

$$J_{L2} = \begin{bmatrix} z_0 \wedge (p_2 - p_0) & z_1 & 0 \\ z_0 & 0 & 0 \end{bmatrix} \quad p_2 = \begin{bmatrix} 0.5 \\ 1 \\ 1 \end{bmatrix} \quad z_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$z_0 \wedge (p_2 - p_0) = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 0.5 & 1 & 1 \end{bmatrix} = -\hat{i} + 0.5\hat{j} + 0\hat{k}$$

$$J_{L2}(q) = \begin{bmatrix} -1 & 1 & 0 \\ 0.5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad v_2 = J_{L2} \cdot \dot{q} = \begin{bmatrix} -1 & 1 & 0 \\ 0.5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{matrix} \dot{p}_2 \\ \omega_2 \end{matrix}$$

As expected, there is only a translation along the x axis

$$J_L \equiv J = \begin{bmatrix} z_0 \wedge (p-p_0) & z_1 & z_2 \\ z_0 & 0 & 0 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} -1 & 1 & 0 \\ 0.5 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

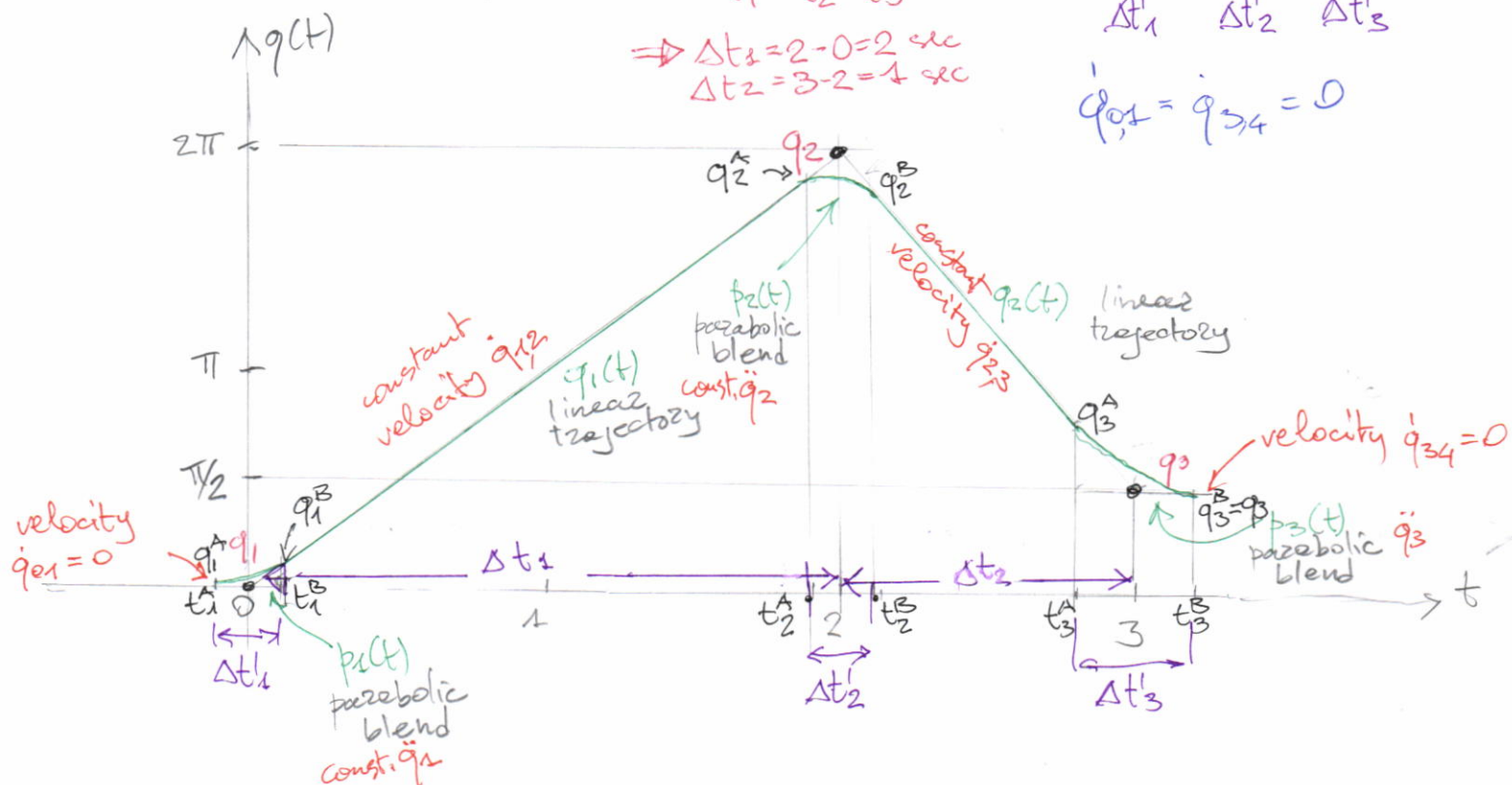
$$v = J \cdot \dot{q} = \begin{bmatrix} -1 & 1 & 0 \\ 0.5 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

velocity of link 3 frame is conditioned by translation along x_0 and z_0 axis due to joint 2 and 3 motion; in such a configuration (q) , joint #2 axis is aligned with x_0 .

B Trajectory through via points

$$q = [0, 2\pi, \pi/2] ; t = [0, 2, 3] ; \Delta t' = [0.2, 0.2, 0.4]$$

$q_1 \quad q_2 \quad q_3$ $t_1 \quad t_2 \quad t_3$ $\Delta t'_1 \quad \Delta t'_2 \quad \Delta t'_3$



Motion laws for linear paths: $q_k(t) = q_k + (q_{k+1} - q_k) \frac{(t - t_k)}{\Delta t_k}$ $k=1, 2$

Motion laws for parabolic blends: $p_k(t) = a_{k0} + a_{k1}t + a_{k2}t^2$ $k=1, 2, 3$

LINEAR PATHS:

$$\begin{cases} q_1(t) = q_1 + (q_2 - q_1) \frac{(t - t_1)}{\Delta t_1} = 2\pi \frac{t}{2} = \pi t & \text{for } t_1^B \leq t \leq t_2^A \\ q_2(t) = q_2 + (q_3 - q_2) \frac{(t - t_2)}{\Delta t_2} = 2\pi + (\pi/2 - 2\pi) \frac{(t - 2)}{1} = -\frac{3}{2}\pi t + 5\pi & \text{for } t_2^B \leq t \leq t_3^A \end{cases}$$

CONSTRAINTS ON VELOCITIES AND ACCELERATIONS:

$$\dot{q}_{k-1,k} = \frac{q_k - q_{k-1}}{\Delta t_{k-1}} \quad \ddot{q}_k = \frac{\dot{q}_{k,k+1} - \dot{q}_{k-1,k}}{\Delta t'_k}$$

Computation of velocities on linear paths and of accelerations on parabolic blends:

VELOCITIES:

ACCELERATIONS:

$$\left\{ \begin{array}{l} \dot{q}_{01} = 0 \\ \dot{q}_{12} = \frac{q_2 - q_1}{\Delta t_1} = \pi \\ \dot{q}_{23} = \frac{q_3 - q_2}{\Delta t_2} = -\frac{3\pi}{2} \\ \dot{q}_{34} = 0 \end{array} \right.$$

$\Rightarrow$

$$\left\{ \begin{array}{l} \ddot{q}_1 = \frac{\dot{q}_{12} - \dot{q}_{01}}{\Delta t_1} = \frac{\pi}{0.2} = 5\pi \\ \ddot{q}_2 = \frac{\dot{q}_{23} - \dot{q}_{12}}{\Delta t_2} = -\frac{25\pi}{2} \\ \ddot{q}_3 = \frac{\dot{q}_{34} - \dot{q}_{23}}{\Delta t_3} = \frac{15\pi}{4} \end{array} \right.$$

CONNECTION POINTS:

$$\left[t_k^A = t_k - \frac{\Delta t_k}{2} ; t_k^B = t_k + \frac{\Delta t_k}{2} \right]$$

$$\left\{ \begin{array}{l} t_1^A = t_1 - \frac{\Delta t_1}{2} = -0.1 \\ t_1^B = t_1 + \frac{\Delta t_1}{2} = 0.1 \end{array} \right. \quad \left\{ \begin{array}{l} q_1^A \equiv q_1 = 0 \\ q_1^B = q_1(t=t_1^B) = 0.1 \times \pi = 0.314 \end{array} \right.$$

$$\left\{ \begin{array}{l} t_2^A = t_2 - \frac{\Delta t_2}{2} = 1.9 \\ t_2^B = 2.1 \end{array} \right. \quad \left\{ \begin{array}{l} q_2^A = q_1(t_2^A) = 1.9 \times \pi = 5.969 \\ q_2^B = q_2(t_2^B) = 5\pi - \frac{3\pi}{2} \times 2.1 = 5.812 \end{array} \right.$$

$$\left\{ \begin{array}{l} t_3^A = 2.8 \\ t_3^B = 3.2 \end{array} \right. \quad \left\{ \begin{array}{l} q_3^A = q_2(t_3^A) = 5\pi - \frac{3\pi}{2} \times 2.8 = 2.513 \\ q_3^B \equiv q_3 = \pi/2 \end{array} \right.$$

PARABOLIC BLENDS:

(determination of a_{10}, a_{11}, a_{12})

$$\left\{ \begin{array}{l} p_1(t) = a_{10} + a_{11}t + a_{12}t^2 \\ \dot{p}_1(t) = a_{11} + 2a_{12}t \\ \ddot{p}_1(t) = 2a_{12} \end{array} \right. \Rightarrow$$

$$\left\{ \begin{array}{l} p_1(t_1^A) = q_1^A \\ \ddot{p}_1(t) = \ddot{q}_1 \\ \dot{p}_1(t_1^A) = 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} a_{10} + a_{11}(-0.1) + a_{12}(-0.1)^2 = 0 \\ 2a_{12} = 5\pi \\ a_{11} + 2a_{12}(-0.1) = 0 \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} a_{10} = 0.08 \\ a_{11} = 1.57 \\ a_{12} = 7.85 \end{array} \right.$$

$$p_1(t) = 0.08 + 1.57t + 7.85t^2 \quad \text{for } t_1^A \leq t \leq t_1^B$$

$$\begin{cases} p_2(t) = a_{20} + a_{21}t + a_{22}t^2 \\ \dot{p}_2(t) = a_{21} + 2a_{22}t \\ \ddot{p}_2(t) = 2a_{22} \end{cases} \Rightarrow \begin{cases} p_2(t_2^A) = q_2^A \\ \ddot{p}_2(t) = \ddot{q}_2 \\ p_2(t_2^B) = q_2^B \end{cases} \Rightarrow \begin{cases} a_{20} + 1.9a_{21} + (1.9)^2a_{22} = 5.96 \\ a_{20} + 2.1a_{21} + (2.1)^2a_{22} = 5.81 \\ 2a_{22} = -\frac{2.5\pi}{2} \end{cases}$$

$$\Rightarrow \begin{cases} a_{20} = 70.85 \\ a_{21} = 77.70 \\ a_{22} = -19.64 \end{cases} \quad p_2(t) = 70.85 + 77.70t - 19.64t^2 \quad \text{for } t_2^A \leq t \leq t_2^B$$

$$\begin{cases} p_3(t) = a_{30} + a_{31}t + a_{32}t^2 \\ \dot{p}_3(t) = a_{31} + 2a_{32}t \\ \ddot{p}_3(t) = 2a_{32} \end{cases} \Rightarrow \begin{cases} p_3(t_3^B) = q_3^B \\ \dot{p}_3(t_3^B) = 0 \\ \ddot{p}_3(t) = \ddot{q}_3 \end{cases} \Rightarrow \begin{cases} a_{30} + a_{31}(3.2) + a_{32}(3.2)^2 = \pi \\ a_{31} + 2a_{32} \times 3.2 = 0 \\ 2a_{32} = \frac{15\pi}{4} \end{cases}$$

$$\Rightarrow \begin{cases} a_{30} = 61.90 \\ a_{31} = -37.70 \\ a_{32} = 5.89 \end{cases} \quad p_3(t) = 61.90 - 37.70t + 5.89t^2 \quad \text{for } t_3^A \leq t \leq t_3^B$$

Motion Law

$$\begin{cases} q(t) = 0.08 + 1.57t + 7.85t^2 & -0.1 \leq t \leq 0.1 \\ q(t) = \pi t & 0.1 \leq t \leq 1.9 \\ q(t) = 70.85 + 77.70t - 19.64t^2 & 1.9 \leq t \leq 2.1 \\ q(t) = 5\pi - \frac{3}{2}\pi t & 2.1 \leq t \leq 2.8 \\ q(t) = 61.90 - 37.70t + 5.89t^2 & 2.8 \leq t \leq 3.2 \end{cases}$$